



Data-driven business process management-based development of Industry 4.0 solutions



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ABSTRACT

Business process management (BPM) supports the management and transformation of organizational operations. This paper provides a structured guideline for improving data-based process development within the BPM life cycle. We show how Industry 4.0-induced tools and models can be integrated within the BPM life cycle to achieve more efficient process excellence and evidence-based decision-making. The paper demonstrates how standards of machine learning (CRISP-ML(Q)), BPM, and tools of design science research can support the redesign phases of Industry 4.0 development. The proposed methodology is carried out on an assembly company, where the proposed improvement steps are investigated by simulation and evaluated by relevant key performance indicators.

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Introduction

The manufacturing sector highly affects the competitiveness of Europe as it accounts for 16% of Europe's GDP. This sector is responsible for 64% of expenditure with regard to Research & Development in the private sector, and 49% of cost related to innovation in Europe [1]. Industry 4.0 (I4.0) is an emerging phenomenon that seeks to provide a networked ecosystem which aims to achieve a more efficient and customer-oriented operation, sustainable competitiveness and smart growth. The digital transformation of manufacturing enterprises should result in more efficient business processes. To support this development, this paper aims to study how the business process management (BPM) of manufacturing companies should focus on developing I4.0-driven technologies and how business process redesign (BPR) can utilize the advanced tools and methods developed in the Fourth Industrial Revolution. Furthermore, the contribution of the design science approach to the BPM life cycle model is explored to highlight innovative and data-driven process development.

The major features of integrating I4.0 into business management are the real-time capability, interoperability, modularity, decentralization and virtualization as well as the horizontal and vertical integration of processes through ICT (Information and Communications Technology) systems [2,3]. This digital

transformation results in Cyber-Physical Systems (CPSs) [4] that have several potentials to improve the efficiency of manufacturing [5].

Businesses can gain a competitive advantage through this data-driven operation, enable innovation and product life cycles to be shortened as well as predict and respond to market demand in time [6]. The realization of data-driven workflow management, both vertically and horizontally, facilitates collaboration between multiple devices - namely, Machine to Machine (M2M), business management systems and production lines - that is Business to Manufacturing - in others words, Business to Business (B2M), and data sharing between business units (B2B) [7].

The adaptation of these changes requires a new perspective and leads to the need to rethink as well as redesign business processes and models [8], for which BPR in the context of design science research (DSR) is effective.

"BPR is an organizational initiative for achieving competitive multi-faceted advantages regarding business processes, in terms of cycle time, quality, cost, customer satisfaction and other critical performance metrics" [9]. DSR contributes to business process redesign through iterative process analysis. DSR is a problem-solving paradigm that supports innovative, knowledge- and data-centric artifacts creation in the form of models, methods, constructs and instantiations [10]. In a I4.0 manufacturing organization context, artifacts are considered to be I4.0 technologies, tools and methods. The concept of DSR can be carried through the whole BPM process life cycle.

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Modeling notations are considered as tools of BPR, such as the Business Process Model and Notation (BPMN), Unified Modeling Language (UML), XML Process Definition Language (XPDL), Petri Net, Integration Definition Method (IDEF0 and IDEF3) and Business Process Execution Language for Web Services (BPEL4WS) [11]. However, in the following, only the BPMN is discussed more precisely as it is mainly for business process modeling and its structure is clearer and easier to use [11]. BPMN creates a standardized bridge over the gap between business process design and process implementation [12]. This technique promotes the automated execution of workflows and provides a comprehensive model, the graphical notion of which is a Business Process Diagram (BPD) [12]. An adequate BPMN model is needed to gain an easily applicable data-centric workflow optimization. The defined processes in BPMN can be further improved and optimized by applying the business process simulation approach [13]. Simulation techniques allow for complex data-intensive processes to be analyzed and diverse scenarios evaluated [14]. However, the potential of digital twins [15] provides a new level of efficiency in terms of adapting and controlling large complex business processes [16].

The purpose of this paper is to develop a methodology to improve data-driven process development, resulting in an evidence-based decision support tool in BPM. It aims to facilitate digital transformation and the conversion of production into flexible manufacturing systems, therefore, improving business agility.

Organizational I4.0 developments require a clear understanding of the 'as-is' state [17]. Maturity models highlight the strengths and weaknesses [18]. Maturity models measure the readiness to adapt digital transformation and data-driven solutions. The adaption of the new technologies and strategic operational processes requires the redesign of the business processes.

The successful realization of the I4.0 concept demands the applications of ICT and automation technologies as well as the related standards and ontologies. Trends in I4.0 define the need for interoperability amongst actors, sensors, and heterogeneous systems [19]. Standards, formal models and ontologies play a significant role in the I4.0 system environment, for which enforcement, linked data should rely on systematically enriched information [20].

Such I4.0 standards describing reference architectures for smart factories are the ISA-95 [21], ISA-S88 [22] or RAMI4.0 [23] models. Standards Ontology describes the characteristics as well as relations between standards and organizations for publishing standards in addition to standardization frameworks in the context of I4.0 [24]. Ontologies can be considered as the next generation of standards for connecting information [20]. To develop interoperability, the Industrial Ontologies Foundry (IOF) works with the governmental, industrial and academic sectors as well as standards organizations by creating a set of core and open reference ontologies covering the entire domain of digital manufacturing [25]. Therefore, semantic technologies are essential in the field of ICT and automation technology (AT) to exploit and connect information between and within companies [26]. The development of I4.0-induced business processes management based on standardized domains can result in a better served and more transparent structure to manage the robustness of the system with a high degree of reliability.

The overall purpose of this paper is to provide a data-driven method for the analysis and development of Industry 4.0 solutions based on business process modeling and simulation.

The proposed methodology is based on the improvement of the BPM life cycle with the integration of a data-driven application. The phases of the BPM life cycle are connected with diverse data-based technologies, tools and methods to facilitate its successful adaptation and efficient operation. The potential in data analysis is highly connected to this concept, therefore, this linkage is highlighted by providing an overview of the integrability of CRISP-DM/ML standards into the BPM life cycle.

This concept is demonstrated through the development project of a Hungarian assembly and engineering company that produces customized and fully autonomous factory units. The illustrative example demonstrates how the redesign of a production line consisting of Computer Numerical Control (CNC) machines can be supported by BPM and BPR tools.

The aforementioned I4.0-induced business process transformation models, methods, and support tools are explored in the following sections.

- *Development of the Business process management (BPM) life cycle using data-driven applications and standards* highlights that the BPM life cycle of I4.0 technologies should intensively utilize data-driven process development tools so that development projects can be efficiently supported by the integration of the standards of data mining (DM) and machine learning (ML) development projects into the BPM life cycle.
- Based on the critical analysis of BPM, I4.0, and CRISP-DM/ML standards as well as methods, a detailed methodological guideline has been formulated for managing business processes in the field of I4.0 in *Methodology - guideline for managing data-driven business processes*.
- Some details related to the applicability of the method are also presented in an illustrative case study in *Lessons learned during the redesign of a CNC production line*.
- Finally, the conclusions and future trends in terms of business data and process management are discussed in *Conclusion*.

Business Process Transformation in the age of Industry 4.0

The realization of digital solutions is a long-term commitment towards the evolutionary path of I4.0 [27]. Existing business processes must be developed within the organization as well as business models [28]. Even though diverse implementation strategies exist for realizing I4.0 within a company, they all seek to achieve transparency, visibility, interconnectivity and efficiency. Ref. [28] defines a road map for implementing I4.0 in terms of enterprises, the major step of which are analysis (understanding the I4.0 concept and the maturity of organization - 'as-is' state), targets (define areas of action, evaluate conditions - redesign processes), realization (prepare decisions and define projects - 'to-be' model and implementation strategy), while the implementation strategy to achieve value-added networks is assessed in Ref. [27]. Its core components are defined as research and innovation, reference architecture and standardization as well as the security of networked systems, which are accompanied by the creation of legal framework conditions.

The six steps for implementing strategic development are shown in Fig. 1. The concept shows how maturity models are able to foster business transformation and allows phases of the BPM life cycle to be linked to digital and technological domains as well as supports the integration of data analysis into the transformation framework [29]. The BPM life cycle is a comprehensive concept which enables the organizational transformation of processes to be managed [30]. Its phases are the following: process identification, process discovery, process analysis, process redesign, process implementation, process monitoring and process control.

The development cycle of implementing strategic changes includes [29]:

1. The digital maturity assessment of an organization, which is an essential step to derive a common understanding of the 'as-is' state of processes (maturity models and process identification).
2. The exploration of possible threats and opportunities is a foundational step towards redesigning our processes (process identification and discovery).

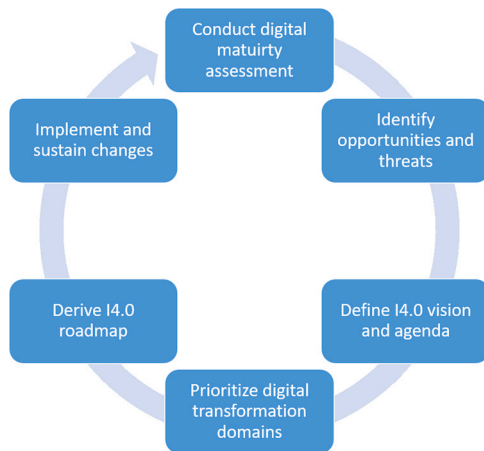


Fig. 1. Recommended steps for implementing strategic development [29].

3. The development of an 'ideal' vision and agenda for creating the 'to-be' goal model of an organization (process analysis and redesign).
4. "Development domains should be prioritized because the creation of Industry 4.0 systems is a complex activity, since the very nature of this concept implies that it affects all business and technical sub-processes" [31] (process analysis and redesign).
5. An I4.0 road map can serve as the 'to-be' model of the redesigned processes, which can facilitate the transformation process coordinating the actions (process redesign).
6. The realization and implementation of the derived I4.0 road map (process implementation and monitoring).

With regard to research projects, information technology-based design science research approach is crucial as it is focusing on developing knowledge and support innovative problem solving through artifacts as models and methods. DSR identifies the problem and its impact, thus reveal potential redesign areas. DSR iterates how would a better artifact (e.g. I4.0 solutions) accomplish in the problematic environment, develops and design a solution, then apply it and measure performance metrics to access effectiveness of the redesigned process. Finally communication of the results is essential for further researches [32]. DSR is a research paradigm with a focus on generating hypotheses on how the future could look ('to-be') and iterates solutions for the redesign. It is initially filtering them to remove hypotheses that are not worth pursuing. "By widening the employment of DSR, the information science discipline can acquire a leading position in the field of practice" [33,34]. The aforementioned concept covers and facilitates BPM as well as the integration of digital technologies, tools and methods for redesigning and implementing I4.0 projects in addition to transformed business processes. The description of the BPM life cycle model and its technology-driven support as well as the possible data-analytic connection to this concept is given in Section 2.1.

Development of the Business process management (BPM) life cycle using data-driven applications and standards

This section discusses the foundations of BPM and the technology alignments to support efficient governance for process-oriented organizations as well as highlights the benefits of utilizing the integration of data-driven applications and standards into the BPM life cycle. *Improvement of the BPM life cycle with the integration of technology alignments* defines how I4.0-driven technologies, tools and methods can be utilized through phases of the BPM life cycle, while *Standards with regard to business analytics - the CRISP-DM/ML framework* provides an insight into the potential integration of data

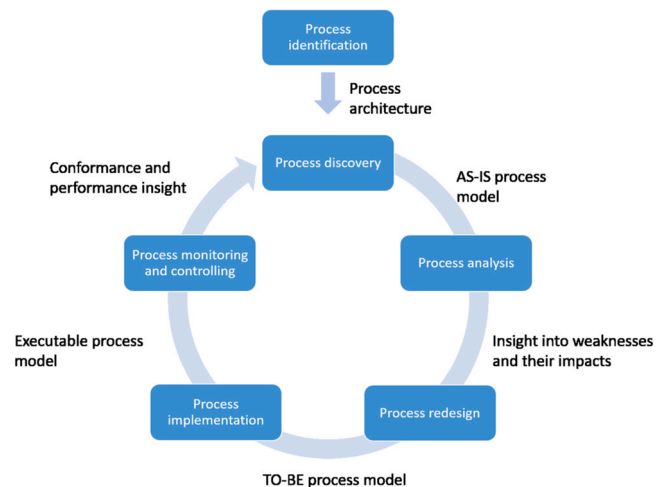


Fig. 2. A model of the Business Process Management (BPM) life cycle.

mining and machine learning standards (CRISP-DM/ML) into the BPM life cycle.

Improvement of the BPM life cycle with the integration of technology alignments

Business processes can be considered combinations of interconnected events, activities and decision points that include many actors and information carriers. Altogether, they contribute to a value-added result for at least one customer [35]. Due to constant digital transformation, managing business processes is no longer a choice but a necessity to remain competitive and productive in the market [36]. Although the role of digital technologies in transforming business processes seems to be under-investigated [37], BPM is a comprehensive system for managing and transforming organizational operations through business process modeling, execution, and evaluation, with agility and operational performance goals [30]. To monitor and continuously develop business processes, BPM utilizes several methods, policies, metrics, management practices, and software tools [36]. In Fig. 2, the model of a BPM life cycle is shown.

Technological developments and digital innovation make it necessary for BPM to be thought through afresh. "Emerging technologies have capabilities to reshape business process management from its traditional version to a more explorative variant" [38]. Ambidextrous BPM mixtures consist of two aspects: exploiting the benefits of existing technologies (i.e., exploitative BPM), while simultaneously exploring the benefits of new IT (i.e. explorative BPM) [39].

The innovation- and data-driven perspectives of transforming business models and processes and the technological tools and methods to support BPM cannot be neglected. "Technology is not the goal, but the instrument and process excellence is the driver to introduce new technologies in the operations" [40].

The digitization of manufacturing enterprises creates the basis for I4.0 principles such as interoperability, decentralization, virtualization, real-time capability, service orientation, and modularity [36]. Organizations seek to be stable, flexible and capable of adapting to changes, while organizational stability provides agility and innovation platforms [41]. Continuous improvement in terms of managing business processes requires the combination of BPM with digital leadership [41], to derive a maturity level that can help to accomplish capabilities for organizational success [42]. In order to achieve real-time capability, cloud computing offers immediate access to hardware resources as well as lowers IT barriers and opens up new ways to innovate [43]. Machine learning standards, tools,

Table 1
Industry 4.0-induced applications, tools and methods supporting the phases of the Business Process Management (BPM) life cycle.

BPM life cycle phase	Definition	Supporting tools	Benefits	References
Process identification	Processes relevant to the problem being addressed are identified, delimited, and inter-related. The outcome of process identification is a new or updated process architecture, which provides an overall picture of the processes in an organization and their relationships with each other.	Maturity models, BPMN, Process mining, Cloud computing, Digital twin	Transparency, Virtualization, Real-time capability, Interoperability	[48–53]
Process discovery	The current state of each of the relevant processes is documented, typically in the form of one or several 'as-is' process models.	Process mining, Cloud computing, Digital twin	Transparency, Virtualization	[51–54]
Process analysis	Issues associated with the 'as-is' process model are identified, documented, and, whenever possible, quantified using performance measures. The output of this phase is a structured collection of issues. These issues are prioritized based on their potential impact and the estimated effort required to resolve them.	Big data analytics, KPIs, Bottleneck analysis, Cloud computing, AI, Process mining, Process model repair	Transparency, Interoperability, Real-time capability, Interconnectivity	[52,53,55–60]
Process redesign	Multiple change options are analyzed and compared in terms of the chosen performance measures. Process redesign and process analysis go hand in hand: as new change options are proposed, they are analyzed using process analysis techniques. Eventually, the most promising change options are retained and combined into a redesigned process. The output of this phase is typically a 'to-be' process model.	Optimization techniques, Standards, Ontologies and semantic models, Simulation and modeling techniques	Transparency, Interoperability, Interconnectivity, Virtualization	[14,61–71]
Process implementation	The changes required to move from the 'as-is' process to the 'to-be' process are prepared and performed. Process implementation covers two aspects: organizational change management and automation. Organizational change management refers to the set of activities required to change the way of participants involved in the process work. Process automation refers to the development and deployment of IT systems (or enhanced versions of existing IT systems) that support the 'to-be' process.	RPA, IoT, CPS, Cloud computing, Digital twin, Cloud manufacturing, Simulation and modeling techniques, Collaborative/Autonomous robots, AI, AR	Interoperability, Interconnectivity, Real-time capability, Virtualization	[72–78]
Process monitoring and controlling	Once the redesigned process is running, relevant data are collected and analyzed to determine how well the process is performing with respect to its performance measures and performance objectives. Bottlenecks, recurrent errors, or deviations with regard to the intended behavior are identified and corrective actions are undertaken. New issues may then arise, in the same or in other processes, which requires the cycle to be repeated on a continuous basis.	IoT, CPS, Cloud computing, MES, Simulation and modeling techniques, AI, AR, Process mining, Digital twin	Interoperability, Interconnectivity, Real-time capability Virtualization	[53,55,77,79,80,78]

methods and approaches can facilitate the automation and optimization of processes as well as support BPM through new algorithms [44]. In the context of BPM, Information Systems (IS) supporting the processes produce the structured data, i.e., logs, which are analyzed with process mining techniques used to discover processes and bottlenecks, check compliance, and propose process improvements [45,46].

Furthermore, a direct link between goals, objectives and strategies regarding key performance indicators (KPIs) is needed [47] to support evidence-based decision-making through the BPM life cycle.

In Table 1, we determine I4.0-induced, data-driven applications, tools, and methods can be integrated into the phases of the BPM life cycle, therefore, facilitate more efficient evidence-based decision-making. The table includes a short definition of the phases of the BPM life cycle based on Ref. [35], then the supporting tools, that enable the operation of the phase to be improved are discussed then the references and examples regarding its benefits are defined.

In the following, the phases of the BPM life cycle are explored with a significant focus on supporting digital technologies, tools and methods.

The **process identification** phase can be supported by organizational maturity models, digital technologies and tools such as process mining, the Business Process Model and Notion (BPMN) and the Digital Twin of an Organization (DTO). Conceptualizing and measuring the maturity of an organization or a process facilitates the understanding of business architectures, therefore, allows the 'as-is' state of processes to be detected [48]. BPMN 2.0 is a standard notation for modeling business processes [81]. The flat control flow perspective, sub-processes, data flows and resources can be integrated within one BPMN diagram, which is interesting from the process miners' point of view [49]. BPMN serves as a representational of modeling processes and opens excellent perspectives for the applicability of process mining. "Discovered process models become available and understandable, and the models can be imported/exported from/to any BPMN modeling tool and executed, process mining techniques can be easily integrated to the existing suites (BPMN serves as an interface in this case). Moreover, BPMN models allow for the combination of different perspectives varying from control flow (including sub-processes) to the perspective of resources" [49]. Process mining is an innovative tool supporting the digital transformation of companies by providing a holistic insight into actual processes and complexities, therefore, allowing the identification of inefficiencies and effort drivers [50]. Process mining techniques should support workflow models such as BPMN [53]. Models extracted from event data provides a purposeful abstraction of reality. Therefore, due to the dynamic nature of processes, process mining should be considered as a continuous process to create a "living and learning process model" [53]. The next level realization of current processes is utilizing digital twins. Digital Twin of an Organization (DTO) serves as a virtual copy of processes, therefore, promotes the understanding of process architectures [52]. A DTO is "a dynamic software model of any organization that relies on operational and/or other data to understand how an organization operationalizes its business model, connects with its current state, responds to changes, deploys resources and delivers expected customer value" [51].

Process discovery can be facilitated by process mining and the perspective of a DTO. Process mining enables businesses to analyze business processes based on digital traces captured in event logs. It is an effective tool to visualize the 'as-is' process flows and check their actual conformance [52]. A DTO provides a virtual interpretation of actual process flows [51], therefore, allows a punctual 'as-is' model of processes to be created. Furthermore, the Business Process Catalog and Classification Scheme (BPCCS) can foster the discovery of business process elements by providing the shared vocabulary of various business process models (e.g. activities, sub-processes, roles)

[82]. "Combining Robotic Process Automation (RPA) with the popular BPM approach poses a useful strategy as the 'as-is' process is optimized first". Subsequently, the RPA procedure is then deployed on the optimized process to reach its full automation potential [54].

The phase of **process analysis** can be improved by process mining, big data analytics, KPIs, artificial intelligence (AI) or diverse analyzing techniques such as bottleneck analysis. Big data management and analytics cannot be neglected in terms of process analysis. Big data analytics allows business processes to be measured and knowledge to be directly translated into decision-making and performance [83]. Big data enables internal processes to be streamlined by reducing the bottlenecks, therefore, increasing the efficiency of processes [59]. Identifying bottlenecks is essential as it affects the operation of the whole system. The bottleneck of a manufacturing system refers to a subsystem that limits the capacity of the entire system. Overall Equipment Effectiveness (OEE) can be used as a KPI that indicates a system's overall capacities and subsystems. However, the lowest OEE may not provide information about bottlenecks because other subsystems can cause them. To overcome this drawback, the standalone OEE (OEE_{sa}) was formed that reveals the true capability of the system, which is partially hidden due to external factors [58]. KPIs measure performance in terms of business goals, typically reflecting the time, cost and quality domain. A direct link exists between KPIs, objectives, and strategies [47]. In terms of evaluating 'as-is' processes, registered KPI values can be verified by process mining to determine deviations from KPI target values, therefore, facilitate the redesign phase [84].

Process mining enables structured data from a process perspective to be analyzed. In some cases, it allows KPIs to be modified in ways that optimize the output to facilitate data-based root cause analysis, which enables errors to be directly connected to relevant process steps as well as identifies unwanted effects [55]. Process mining techniques enable to observe behavior (e.g. event logs) to modeled behavior (e.g. BPMN model), therefore, detect and diagnose differences between observed and modeled behavior [60]. Process Model Repair uses a reference process model and an event log as an input. If the model cannot fully explain the observed behavior, these parts of the model which are not consistent with the observations can be repaired using this technique [60]. Furthermore, the constantly improving technology fosters the continuous development of business processes; thus, business process redesign can be considered as a process enhancement operation (process model repair). Process mining can be further aided by AI, which offers possibilities in predictive process mining [55]. Predictive process mining plays a significant role in business scenarios as it enables the next activity of a running process to be identified in advance. Therefore, it can foster "optimal management of resources and promptly trigger remedial operations to be carried out" [57]. The data gathered about the execution of the processes provides information about decision patterns, performance, or control-flow provides, which information can be utilized in simulation models to perform a new complex process model [53]. The comprehensive twin of all process activities and complexities provides the strategic management of an organization with insights based on facts. Strategic directions such as digital transformation programs become measurable and thus easier to manage [52].

The **process redesign** phase of the BPM life cycle can be improved by I4.0-induced technologies, models and tools, in addition to optimization techniques, standards, ontologies and semantic models, as well as simulation and modeling techniques. A major aspect of I4.0 is to provide a coherent approach for adapting semantic communication between intelligent systems. To redesigning businesses according to this approach, a communication model is needed for which ontologies can provide a solution by formalizing the knowledge of smart manufacturing in an interoperable way. "An unambiguous, semantic-based knowledge representation of

concepts for smart manufacturing domain is required to ensure a coherent and effective human-robot collaboration" [62]. Industry practitioners can use these ontologies to conceptualize implementation scenarios [63]. Indeed, as every scenario considered within the framework of I4.0 includes different entities which communicate and cooperate, the main role of the presented ontological standard is to facilitate that exchange [62]. According to the domains of I4.0, Factory 4.0 or smart manufacturing, both the perspectives of business and production are involved. "As I4.0 relies heavily on robotic agents which have to evolve and perform the main operations in smart manufacturing environment and which are solicited to communicate with human operators, customers, or with diverse distributed partners. The standardization of knowledge representation is a key element facing I4.0 development and is required to be addressed quickly and efficiently to avoid accumulated difficulties at later stages of the development" [62]. Therefore, ontological standardization regarding I4.0 mainly relies on IEEE 1872–2015 - Standard Ontologies for Robotics and Automation [64] which can be extended by specific ontologies. These ontologies, for example, the Core Ontology for Robotics and Automation (CORA), Ontology for Autonomous Robotics (ROA), Ontology for Robotic Architecture (ORArch) or Ontology for Industry 4.0 (O4I4). During process redesign, the architecture of the 'to-be' process model is developed, which in the I4.0 environment cannot neglect the connection between hardware and software. This viewpoint can be supported by ORArch that elaborates on how hardware and software can be represented together in mixed architecture descriptions [62], while O4I4 is a business-focused ontology [62].

Due to the dynamic nature of the business environment, changes in workflow models of processes are becoming increasingly common. "Process mining application provides a visual and fact-based proof for automation capabilities and enables prioritization of activities" [79]. Goal-oriented ('to-be') dynamic workflows seek to provide flexibility concerning the execution of business processes. "BPMN or similar languages must continue to be the primary instruments for defining a workflow" [65]. In general, optimization procedures still do not support full automation, hence automated optimization solutions are in demand, covering the monitoring, identification and modification of process structures [66]. For modeling processes in Cyber-Physical Systems (CPS), the BPMN4CPS model has been proposed [71].

The extension of BPMN is based on the simulation of Cyber-Physical Production Systems (CPPS) in terms of reconfigurability, adaptability and reliability. This approach relies on the principles and standards of Model-Driven Architecture (MDA) to improve automation and customization [70]. Ref. [14] presented a largely automated business process modeling and simulation approach based on BPMN. This approach "first annotates the BPMN model with what is necessary to make the model executable (i.e., performance parameters, execution resources, and expected workload), and then automatically maps the annotated model into simulation code ready to be executed" [14]. Furthermore, the effective implementation of production goals can be achieved by the combination of manufacturing process models. "To accomplish this, a workflow control engine needs to be designed and standardized respectively" in IIoT Systems that apply the Arrowhead Framework [69]. IIoT Systems that apply the Arrowhead Framework addresses how BPMN and Colored Petri Nets (CPN) [85] can be implemented in the Arrowhead Framework, which is an IIoT (Industrial IoT) framework that dynamically and flexibly supports automated manufacturing processes following I4.0 expectations [68]. Furthermore, Robotic Process Automation (RPA) is an emerging technology to automate business processes that are driven by user interaction with software systems [86]. The integration of RPA into BPM allows their technologies to be connected and systematic methods combined [87].

The **process implementation** phase of the BPM life cycle has much potential to be supported by developing technologies and models. In addition to RPA, emerging phenomena include collaborative and autonomous robots, cloud manufacturing, IoT, Digital Twins, simulation and modeling, and the umbrella term of AI. Cloud manufacturing is a networked, intelligent manufacturing model that integrates IoT, the Semantic Web, cloud computing, high-performance computing and informatized manufacturing to derive a reliable, highly efficient, and quality service for the whole manufacturing life cycle [72]. Cloud manufacturing should be integrated into the BPM life cycle to support business process evolution and adaptation as well as automate this procedure using the Business Process Cataloging and Classification System (BPCCS) [73]. In terms of improving the design and implementation of BPM, a sensor-based IoT network technology can aid this process [74]. IIoT further empowers companies to adopt data-driven strategies and solutions. Being flexible and adaptive to changes in business processes, this paradigm transform industries into cyber-physical production systems [75]. Furthermore, the process-based framework of self-adaptive workflows for CPS ensures agility and autonomy [76]. Implementing Enterprise Resource Planning (ERP) in an organization, the phases of the BPM life cycle must be continuously managed and monitored. According to an Indonesian case study, leading companies "implementing ERP for more than five years, obtained high scores for BPM implementation. They perform well in terms of process identification, implementation, monitoring and control, but are weak in process discovery and redesign, mainly because they do not optimally use specific tools for process modeling and there is a lack of process governance" [77]. The implementation of Robotic Process Automation (RPA) provides a virtual workforce to automate manual, repetitive and error-prone tasks, the solutions of which will lead to the concept of socio-cyber-physical work systems [88]. The digital replica (digital twin) of the system can be efficient to test and continuously monitor the changes in business processes. Digital twin (DT) is an emerging technology used in intelligent manufacturing, which is the digital copy of a physical system and captures its attributes and behaviors in real-time and predicts (simulate) system failures. The implementation of digital twins enables us to gain an understanding of how the physical system would operate after implementing changes due to business process redesign [78].

The **process monitoring and controlling** of the whole life cycle is essential for the successful operation. This phase of the BPM model spans the whole life cycle, not only the monitoring of the results but continuous data collection and model analysis. These activities can be highly supported by data-driven technologies, AI and ML functions, CPS and simulation, as well as process mining or digital twin technologies. Process mining can be used as a "monitoring tool in production that allows data-based, fast decisions. These can help to reduce rework, improve the quality of our products, and reduce production costs" [55]. Process mining further allows the visual and fact-based proof of automation capabilities, prioritizes activities, and supports process automation. Successful process automation demands knowledge about automation potential, effective training of the robots and continuous monitoring of their performance. [79]. Business process automation must be continuously managed and monitored [77]. Regarding manufacturing companies, trends have shifted towards individualized mass production, which presents challenges in achieving interoperability between physical and digital systems concerning the intelligent organization of resources. To manage this, a digital twin-driven Manufacturing Cyber-Physical System (MCPS) for parallel controlling of a smart workshop under a mass individualization paradigm can be a solution [80]. The continuous monitoring and prediction of production systems are critical features for which intelligent manufacturing systems provide a real-time solution resulting in lower

cost, increased flexibility, higher productivity, and better quality [89]. The extension of DT is the Hybrid Digital Twin (HT), which “recognize, forecast and communicate less optimal (but predictable) behavior of the physical counterpart well before such behavior occurs. An HT integrates data from various sources (e.g., sensors, databases, simulations) with the DT models and applies AI analytics techniques to achieve higher predictive capabilities while at the same time optimizing, monitoring and controlling the behavior of the physical system. An HT is typically materialized as a set of interconnected models, achieving symbiosis among the DT models” [78]. Beyond DT and HT, the concept of cognitive digital twin (CT) encompasses some forms of cognitive capabilities, which enable improved data-, knowledge- and experience-based automatic decision making. CT overarch the whole life cycle of the system [78].

Standards with regard to business analytics - the CRISP-DM/ML framework

A vast amount of heterogeneous data is generated during organizational operation, enabling a business insight at a higher level to be gained. Big data analytical techniques must be implemented directly into business processes to successfully recognize, analyze and redesign organizational processes as well as outperform competitors [90]. According to a report by Gartner, organizations are motivated to use data mining, AI and ML to improve and automate tasks. Furthermore, they are expected to double the number of machine learning projects within a year [91]. Data mining and machine learning aim to identify patterns and relationships hidden in a vast amount of heterogeneous data to support the decision-making process. However, the major differences between DM and ML are hidden in the time frame and level of automation. Since DM identify patterns on existing data, while ML predicts future outcomes on the ‘pre-existing’ data due to the initially given rules, and the learning and refining are automatic without human intervention [92]. Businesses can improve their data mining analytics with machine learning algorithms to improve efficiency [93]. In many cases, organizations rely on models closely related to ML, such as the Cross-Industry Standard Process for Data Mining (CRISP-DM), grounded on industrial data mining experience. However, CRISP-DM has two significant drawbacks as “not cover the application scenario where an ML model is maintained as an application” and “lacks guidance on quality assurance methodology” [94]. Regarding Ref. [94], a Cross-Industry Standard Process for Machine Learning is defined using quality assurance methodology (CRISP-ML(Q)). The methodology defines quality as it is “not only defined by the product’s fitness for its purpose but the quality of the task executions in any phase during the development of an ML application” [94]. It extends the CRISP-DM principles of ML purposes as it focuses on “technical tasks needed to produce evidence that the development process of a given ML application is of sufficient quality to warrant the adoption into business processes”. The CRISP-ML model is shown in Fig. 3 and consists of six phases: business and data understanding, data preparation, modeling, evaluation, deployment and monitoring, and maintenance. The sequence of these phases can vary and the arrows indicate the most frequent and important dependencies between phases. Iteration and the revisiting of previous steps are needed until success is achieved or criteria are met [94].

Standard processes during business process management are needed, and the increase in the number of I4.0 applications enables a highly automated environment with machine learning subsidies to be created. Therefore, we discuss the CRISP-ML(Q) integration into the BPM life cycle to derive a methodology suitable for industrial parties to manage and support business processes in the I4.0 environment efficiently. Table 2 describes that the key features of

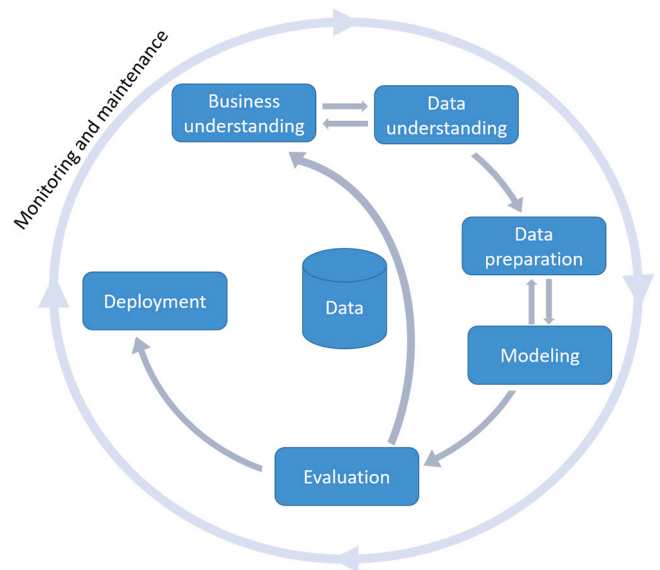


Fig. 3. The process model of CRISP-DM extended by an overarching monitoring and maintenance phase.

CRISP-ML(Q) can be fitted into the phases of the BPM life cycle well as defines the outputs of each phases [95].

The management of business processes by supporting digitization and automated processes via AI has great potential for organizations. The consistent redesign of processes to implement the I4.0 environment is critical, as is the opportunity for companies to use their data and make intelligent, profitable and real-time decisions [44]. ML-related standards, tools, methods, and approaches promote process automation and optimization as well as facilitate BPM through new algorithms and independent learning through continuous data analysis [44]. Automated optimization solutions are required, covering monitoring, identification and modification of the process structure [66]. A crucial element of the redesign of business processes according to the concept I4.0, the standardized knowledge representation [62]. In terms of I4.0 and the intelligent manufacturing environment, ontological standardization relies highly on robotics and automation to facilitate information exchange for communication and cooperation.

Industrial data mining and ML-related process standards (CRISP-DM/ML) are comprehensive process models for data mining/machine learning that provide a foundation for developing a specialized process model ('to-be') and define steps to achieve it [94,96]. These steps are highly interrelated to the phases of the BPM life cycle, as is highlighted in Fig. 3. Integrating the constantly developing technologies and diverse standards into the BPM model facilitates I4.0 fitting as well as highly flexible, interconnected and agile business processes. Furthermore, BPMN models describing both 'as-is' and 'to-be' process models ensure transparency. Simulation models of the processes support optimization by revealing bottlenecks and validating structural efficiency. Therefore, the digital twin of an organization based on the digital copy of a physical asset allows real-time data of the current process to be assessed and collected, information of the interaction with the environment to be derived and information traces captured.

Methodology - guideline for managing data-driven business processes

This section discusses the methodology for developing business process management. This paper provides a guideline for realizing I4.0-driven business processes by utilizing technological

Table 2

Intertwining features between CRISP-ML(Q) and the phases of the BPM life cycle ensuring the suitability of integration.

CRISP-ML (Q) phases	BPM life cycle phases	Features	Outputs
Business understanding	Process identification	Creating an overall process architecture, defining business scope and objective.	Determined business background and objectives.
	Process discovery	Defining 'as-is' state to assure feasibility.	Assessed situation, determined data mining/ machine learning goals and success criteria. Produced project plan, initial assessment of tools and techniques.
Data understanding	Process analysis	Collecting and verifying the data quality regarding the structured issues and, finally deciding upon whether the project should be committed.	Collected initial data, described, explored and verified data.
Data preparation	Process analysis	Producing a data set for the modeling/design phase.	Created dataset and its description (selected, cleaned, constructed, integrated, formatted data).
Modeling	Process redesign	Creating models that satisfy the given constraints and requirements ('to-be' process model).	Selected modeling technique, generated test design, built and assessed model.
Evaluation	Process analysis	Design and analysis are intertwined and as new change options are proposed, they are analyzed using process analysis techniques. The performance, robustness and explainability must be evaluated.	Evaluated results, reviewed process.
Deployment	Process redesign		Determined further steps.
	Process implementation	Preparations to move from the 'as-is' process to the 'to-be' process are made and the transformation is carried out. Before rolling out a model to all, it is best practice to deploy it first to a small subset and evaluate its behavior in a real-world environment (called canary deployment).	Planned deployment, produced final report and reviewed project.
Monitoring and maintenance	Process monitoring and control	Monitoring and maintenance processes to assure quality performance and identify corrective actions. The model has to adapt to changes in the environment.	Planned monitoring and maintenance plan.

developments. Based on the previously presented detailed analysis, this section outlines how and at which stage of a project emerging technologies and data-driven decision support tools can be utilized to endure process efficiency and rebound business agility.

We suggest following the BPM life cycle (Fig. 2) through innovative I4.0 projects which require business processes to be redesigned through iterative, knowledge- and data-centered process analysis, following the design science research paradigm. BPM is a comprehensive system to transform business operations through modeling, execution and evaluation. Its life cycle model provides a transparent reference 'step by step' model to transform processes successfully. To ensure business agility, competitiveness and smart growth, organizations must accomplish a transparent, interconnected, real-time and virtual data-based workflow management. These features can be derived by utilizing the potential of, for example, process mining, IoT, BPMN models, ontologies and semantic models, robotic process automation, digital twins and simulation models. These technological alignments are categorized into the phases of the BPM life cycle outlined in Table 1. Furthermore, we suggest integrating business analytical standards into the BPM life cycle, such as CRISP-DM/ML. Table 2 defines the connectivity features between the models. Businesses' vast amount of heterogeneous data must be managed to continuously recognize, analyze, improve, and implement changes within business processes. Since data mining and machine learning projects closely related to AI tend to increase in an industrial concept, supporting cross-industry standards for machine learning and data mining are essential. Especially in manufacturing, which is of the most promising area to successfully apply digital twin for maintenance and operation monitoring and optimization purposes [97]. Table 3 presents a higher level of integrability of the concept mentioned above related to how phases of the BPM life cycle, as well as data-driven standards and ontologies, are related at different levels of the organizational hierarchy (business, process, implementation). The BPM life cycle and CRISP-ML model is broken down according to organizational hierarchy levels. This point of view supports the comprehensive understanding of which stage of the project belongs to which level of organizational operation.

Lessons learned during the redesign of a CNC production line

This section presents a case study in which a BPM life cycle and I4.0-driven tools are applied to redesign a production line of CNC machines. The case study is of a Hungarian assembly and engineering company oriented in Industry 4.0 technology applications. The project aims to redesign the production line of CNC machines into a fully automated 'smart' factory unit and replace the current product tracking system with a flexible manufacturing execution system (MES). This case of redesigning the business process should significantly build on data-driven and I4.0-induced developments, tools and methods as the design of the planned automated factory unit and integration of the manufacturing execution system into the current phases require the detailed analysis and redesign of the business processes.

Application of the proposed methodology

In this section, the details of the guideline proposed in *Methodology - guideline for managing data-driven business processes* are presented.

The purpose of this demonstration is twofold: 1) the development project through the phases of the BPM life cycle and its decision support tools, then 2) to create the BPMN-based simulation model of the 'to-be' model to validate its structure as well as examine and analyze its processes. Simulation of the process is used as a tool to validate and continuously monitor the efficient operation of the tasks as well as gather real-time information traces and data sets. Fig. 4 shows the workflow of redesigning business processes while considering business analytic (CRISP-DM/ML) aspects at a Hungarian assembly company. The workflow follows the data-based BPM life cycle steps and indicates tools and methods applied at each step.

The current processes were identified through observations, in-depth interviews with employees, organizational databases and enabled to gain a comprehensive understanding of business background and objectives. Processes were discovered by analyzing log files, namely registries of events, processes, messages and

Table 3
The intertwining of the BPM life cycle and standard models at the levels of organizational hierarchy.

Hierarchical levels	Definition	BPM life cycle phases	CRISP-DM/ML phases	Supporting standards
Business	Used by executives and senior business managers to organize their overall understanding, evaluation, and management of a business's performance.	Process identification, Process discovery	Business understanding	BPMN, UML AD, B2MML, OPC UA, IEC 62541, BPDML, OI40, CDD
Process	Used by managers, employees, business analysts, and human performance analysts to change how processes work.	Process analysis Process monitoring and controlling Process redesign Process monitoring and controlling	Data understanding Monitoring and maintenance Data preparation, Evaluation Modeling Monitoring and maintenance	BPMN, ISO/PRF TR 24464, ALPS, ADACOR, BPQL, OPC UA, IEC 62541, BPMML, BPDML, CDD, IEEE 1872–2015 standard
Implementation	Technology used to implement processes according to a context defined in upper levels. It can also be used to develop job descriptions or hire new collaborators.	Process implementation Process monitoring and controlling	Deployment Monitoring and maintenance	AutomationML, FDI, EDDL, AASX, WSBPEL, EDI, ORArch

communications, that were utilized to recreate the proper structure and attributes of processes. As soon as we see how the company works, improvement ideas undoubtedly will show up. BPMN modeling helps to understand the answers to the questions like: How do they work? Why do they work based on this strategy? How can it be improved?. In order to graphically reconstruct the 'as-is' state, a BPMN model of business processes is built to ensure transparency and support the identification and discovery phases and bottlenecks. The data-based analysis of the current process model was carried out by process- and data mining, analysis of KPIs and bottleneck, and ANOVA analysis to reveal weaknesses and possible development areas. Examination of the extracted data and log files enabled to define bottlenecks to be identified. The redesign of business processes was carried out using BPMN modeling for the desired automated manufacturing process. A BPMN model of the 'to-be' improved the current processes, and the automated factory unit ensured its transparency and ability to be simulated (deployed). The improvements based on the KPIs can be evaluated similarly to Ref. [98]. As the application study is based on a continuous development project at the Hungarian assembly company, there were no opportunities to implement the 'to-be' model in a traditional (physical) way. Therefore, a simulation model was developed from one hand to previously assess how the changes would affect the outcome and other processes and reveal further potential bottlenecks. The simulation of the business process model includes information about time periods, executors and resources to analyze processes and reveal possible bottlenecks. A simulation technique is used to find the potential flaws in the current processes and to realize/optimize the design of the process improvements/experiments [99]. The developed simulation model enables the process structure and allocations to be validated, thereby ensuring the implementation phase is efficient.

The simulation environment is implemented in a Python environment with the package called Casymda [100]. Casymda enables SimPy [101] discrete simulation models to be created with the help of BPMN designed using Camunda Modeler [102]. Camunda Modeler helps to ease the creation of simulation models containing more complex processes. By using a defined subset of available BPMN elements combined with naming conventions, model graphs can be created to derive corresponding Python modules from base templates. The model graph is used to define the instantiation of model objects and predecessor/successor relationships between these objects. The initialization of work-in-progress (WIP) in the simulation model is realized using entry points of the model elements. It accounts for seized resources and the processing time that has already passed following the start of the simulation by comparing an entity's entry timestamp to the starting time of the simulation. The entities follow their normal processing flow [103].

The constant monitoring and control function is implemented by detecting information traces through process mining. In order to evaluate the results, indicators were defined concerning, e.g. time, cost, performance, utilization. Four types of performance indicators were used: the mean waiting time of the entities before being processed by the resources, the values of stock waiting to be processed, capacity, and utilization of the resources.

It should be noted that given the demands of continuous improvement, this workflow should return to the first stage.

Results and discussion

Details of BPMN model development

A simplified BPMN model of the processes is shown in Fig. 5. The production processes consist of four segments, which are described below:

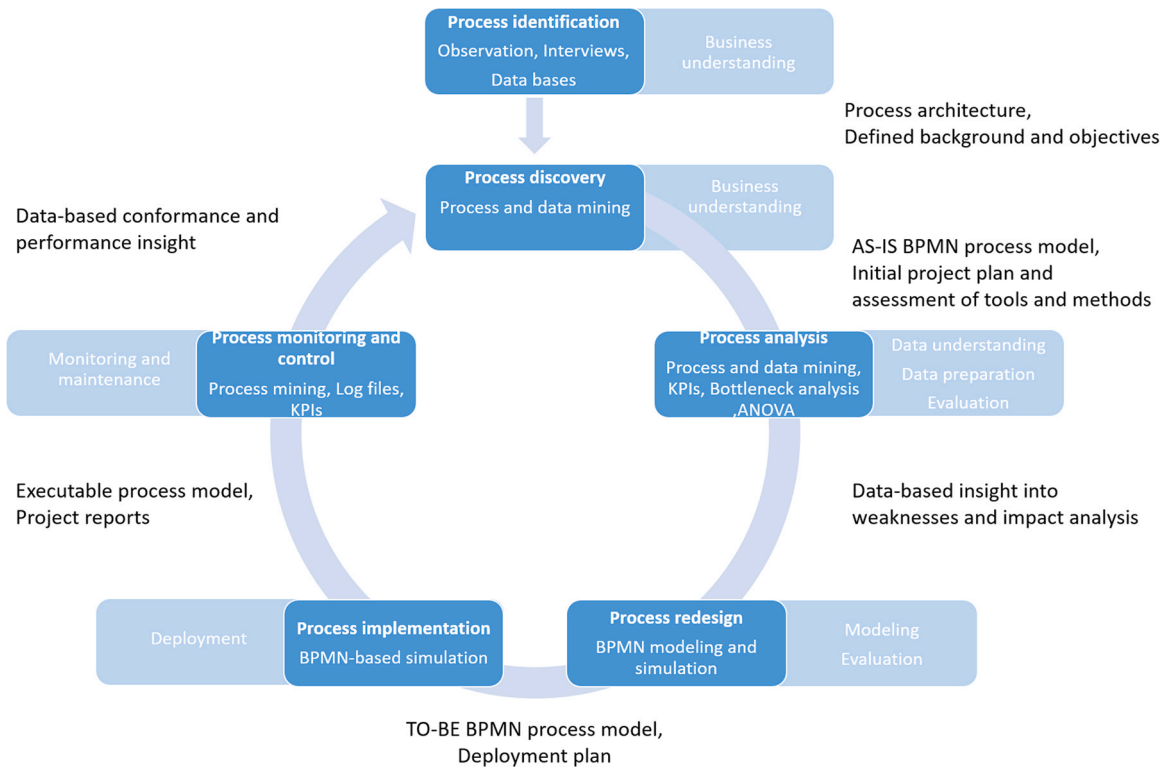


Fig. 4. Workflow of redesigning business processes at a Hungarian assembly company.

- **Technologist work:** One technologist operator makes decisions about the processes to be performed by the ordered product. This operator's tasks are presented in Fig. 5 from P1 to P11 and from P15 to P19. Typically, one technologist operator works during each shift.
- **Programming:** There are programmers who write CAM programs for CNC milling. Their processes are described in Fig. 5 from P20 to P23. Usually, five programmers work during each shift.
- **Cutting plant:** In the cutting plant, the raw materials are cut to the desired dimensions for CNC milling. These processes are

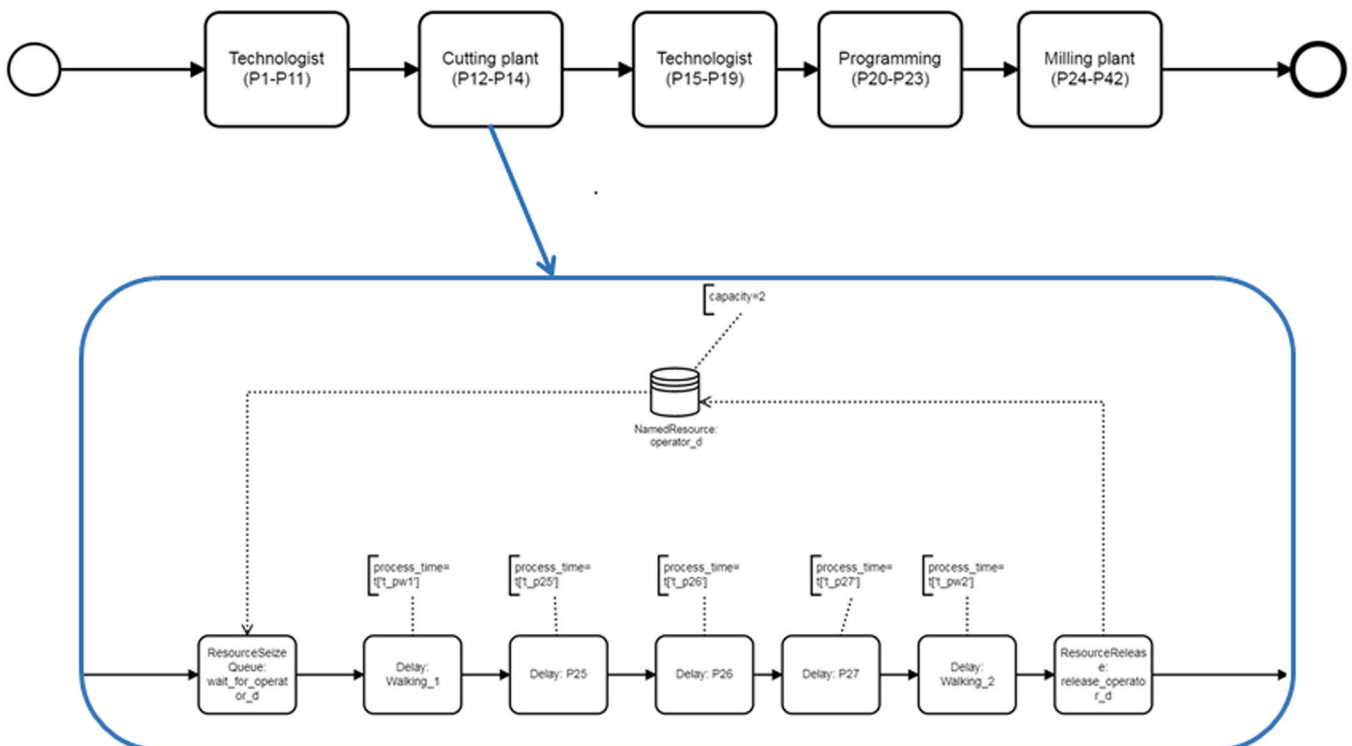


Fig. 5. Process overview.

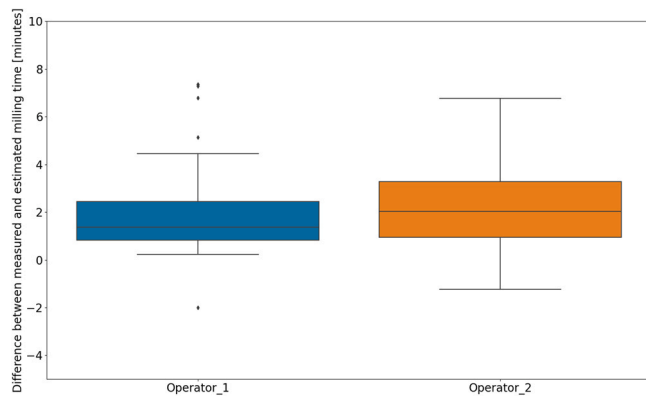


Fig. 6. The box-plots of the difference times between the measured and estimated milling times by the operators.

described in Fig. 5 from P12 to P14, where the walking processes are also indicated. On average, two operators work during each shift.

- **Milling plant:** CNC milling operators handle the CNC machines. Their processes are described in Fig. 5 from P24 to P42 besides the walking processes. Typically, six operators work during each shift and are individually assigned to work on one of the eight CNC machines available.

An overview of the processes is presented in Fig. 5, where the detailed structure of the cutting plant is also presented. The technologist starts the process, who decides about the appropriate steps to perform to get the desired product. Then the cutting machine operators prepare the raw materials for CNC milling. Subsequently, as the exact dimensions of the raw material to be cut are known, the programmer or the CNC operators write the CAM programs to be used for milling. Currently, 30% of the CAM programs are made by CNC operators at the manufacturing plant. As soon as the CAM program is available for production, the CNC operators can commence their work.

As the processes concerning the third stage of the workflow have been thought through, many improvements can be recognized. BPMN-modeling, so the understanding of the 'as-is' state in collaboration with the company's process engineers help a lot in figuring out the possibilities for process improvement. These activities help to understand the reasons behind their working and operating strategy, and they help to identify the bottlenecks. On the other hand, BPMN-based simulators can provide a systematic solution to identify the bottlenecks of the system [99]. In our case, the following improvements are investigated by evaluating different KPIs:

- **First case:** We believe the CNC operators should not spend their valuable time writing CAM programs since this would decrease the utilization of the CNC machines. Programmers should write all the CAM programs.
- **Second case:** While the CNC machine is being operated, the CNC operators can handle other production tasks using other available CNC machines instead of being idle.
- **Third case:** The third possibility is to decrease the non-value added activities of the operators, so AGVs (automatic guided vehicles) can transport raw materials and products instead of the operators. Furthermore, the virtualization of the drawings could be considered, so sheets of paper do not need to be handled.

Exploratory data analysis and prediction model development

An experimental test environment has been installed in the CNC milling plant, which the process engineers consider as a "quasi" smart unit. It practically means a process improved unit, where the

CNC milling operators do not develop CAM programs, and every necessary raw material and tool is immediately available.

To show the difference between the systematic and unsystematic operation, the utilization of the CNC milling machine increased by 30% in the test environment. It leads to the fact that there is a huge potential in the performance improvement at the investigated company, making the goal much more attractive.

However, this process improvement on the whole plant is not easy since it is necessary to explore all the variables affecting the utility and performance. It is necessary to predict how these variables affect the performance indicators. Moreover, the best possible schedule can be reached only with accurate models. For this purpose, we investigated the data provided from the test environment, and we present how the real CNC milling times can be predicted based on the estimated CNC milling times found in CAM files.

We performed an ANOVA test to justify that the difference between the measured and estimated CNC milling times is independent of the operator or not. The null hypothesis is that all difference means are equal, which is rejected since the F-score is 118.6 while the critical value is 3.7. It means that the differences are not independent of the operator, which result is supported too with the box-plots shown in Fig. 6. It is worth noticing that the difference is positive, so the measured milling times are higher than the estimated times.

In the following, we investigated how the difference between the measured and estimated milling times varies with the estimated milling times and with the number of the required tools for milling, as it is presented in Fig. 7. As can be seen, the residuals increase both with the estimated milling times and the number of required tools. Hence both variables are considered in the prediction. Since the residuals are not independent of the operators, two linear models were developed to predict milling times. However, the coefficient of the variable presenting the number of tools in case of *Operator_1* is 0.01 and its p-value is 0.8 (> 0.05), while in case of *Operator_2* the coefficient is 0.3 and its p-value is 0.02 (< 0.05). It means that the number of required tools affected only the milling time in the case of *Operator_2*.

An applicable result for the company is that the measured milling times are not independent of the number of the required tools. It comes from the fact that the tool shelf does not have enough slots to store all the necessary tools, and changing the tools is an additional non-value activity. Therefore, the company will change the shelf to a bigger one and eliminating this unnecessary activity.

On the other hand, using the developed prediction models, the root mean squared error (RMSE) decreased from 2.5 min to 1.4 min in the case of *Operator_1*, and it decreased from 4.5 min to 1.9 min in the case of *Operator_2*, so the activity times of the milling can be predicted more accurately.

We presented a small part of model development, which process accompanies the whole business process improvement. CRISP-ML supports and helps to perform the process development more accurately and systematically.

Details of the simulation and process mining studies

Following the implementation of the suggested improvements, the 'to-be' BPMN model is built. The built BPMN models can be used to simulate the different variations concerning the operations using Casymda and the operation structures evaluated by calculating the relevant KPIs concerning the fourth stage of the workflow.

The first scenario is simulated by directing all the entities directly to the programmers. The second case is simulated by queuing the appropriate resources in the proper order. In the simulation, the entities are first queue CNC machine as a resource. Then, if available, an operator is queued to perform the necessary processes. Then if CNC milling is the process to be performed (P36), the operator is released to undertake other processes. If the milling process is

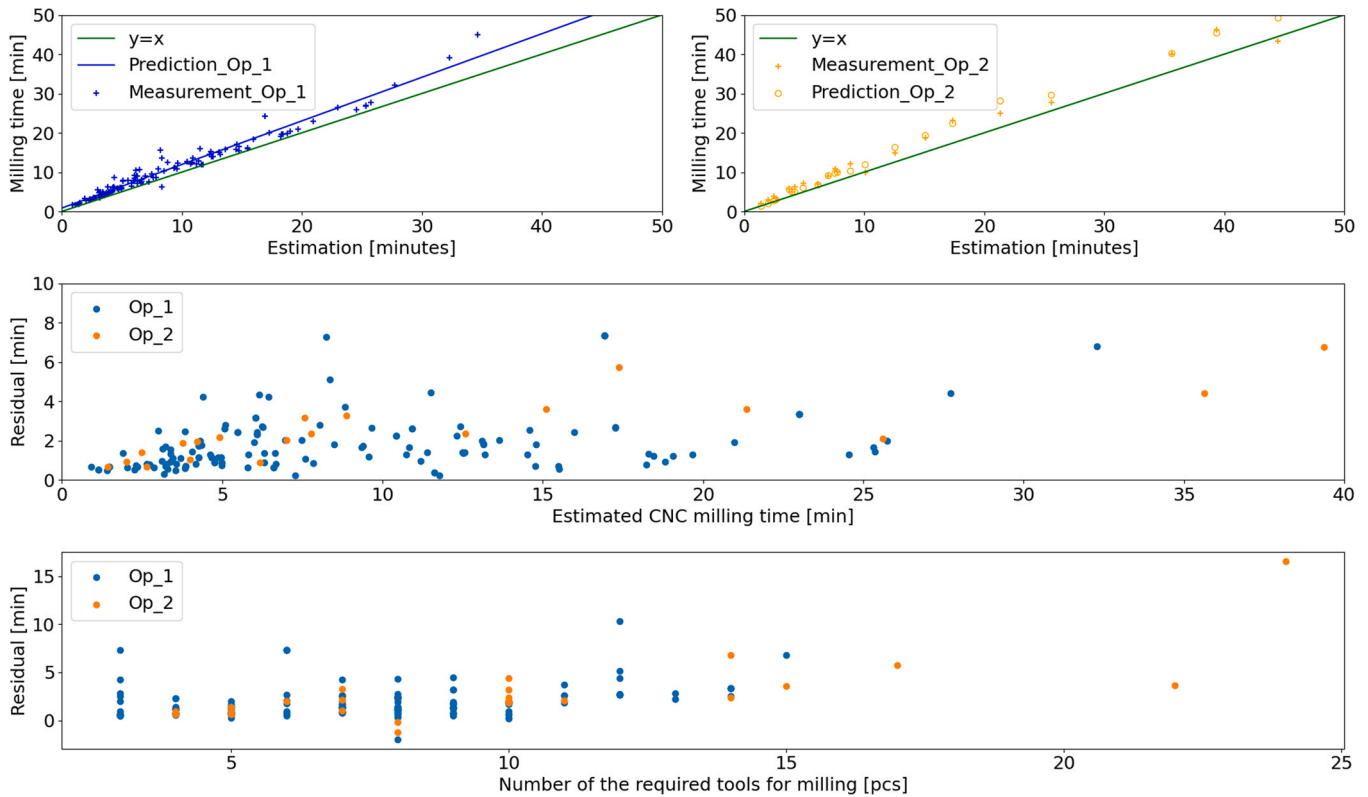


Fig. 7. Analysis and prediction of milling times.

conducted, an operator is required to continue the manufacturing process. The third case is simulated by adjusting the walking time to zero minutes. In this case, it is assumed that an AGV is always available to transport materials, and the transportation processes do not hinder other processes.

Process mining techniques can be applied to validate and verify the developed process model and simulator by analyzing a log file generated with the developed simulator. Process mining is a powerful tool to verify that the obtained results and analysis are consistent with the present 'as-is' and the desired 'to-be' systems. The log file generated from the simulation holds information about the events, processes and communication messages that enable the simulated process to be reconstructed and its operations investigated.

Concerning the sixth stage, we have presented the investigated KPIs in each case, and it can be seen that all the suggestions improve the performance of the manufacturing unit (Fig. 8).

By evaluating step-by-step the KPIs, we can gain valuable information about the processes and their structures. In the first case, all CAM programs are written by the CAM programmers resulting in a slight increase in *KPI2*, therefore, an increase in the quantity produced per shift. As a consequence, the average waiting time of entities (*KPI1*) to be processed by CAM programmers increased, so the resource utilization (*KPI3*) and the stock quantity (*KPI4*). The average waiting time is about 45 min, the average stock is eight pieces, and the degree of utilization per shift is 50%. In the second case, the milling operators can handle more CNC machines, which would suggest a significant increase in capacity. The quantity produced per shift rose from 25 to 35 to be exact. As shown from *KPI1*, the operators are not subjected to significant waiting times, unlike the CNC machines, which means that the number of CNC machines is the relevant constraint. In the third case, the materials are transported by an AGV robot and all information is visualized. It increases the production capacity slightly and reduces the waste time so the cutting operators can finish their tasks earlier, as seen from *KPI4*.

However, this leads to an incremental change in *KPI1* and *KPI4* at CAM programmers. Following the suggested process improvements, we can increase the production capacity, which our simulation has demonstrated.

Evaluation of the results

The evaluation of the simulation results is structured according to the six stages of the workflow shown in Fig. 4.

1. **Process identification:** The combination of the observation method, in-depth interviews and the analysis of databases (product tracking system and CAM files) allowed a common understanding of the current processes as well as the structure and operation of the automated factory unit to be gained.
2. **Process discovery:** The BPMN graphical notion of the 'as-is' model brings about a transparent visualization and logical structure to the concept.
3. **Process analysis:** The extracted data is analyzed by data and process mining tools. The impact of operator on the CNC milling times was analyzed by ANOVA. The model was optimized according to the defined KPIs (e.g. time, cost, utilization).
4. **Process redesign:** The BPMN model of the 'to-be' model is built to ensure transparency. The process model supported the specification of the MES and defined which characteristics we aimed to measure.
5. **Process implementation:** The BPMN-based simulation model of the 'to-be' stage is built. The simulation model is based on BPMN, and parameters concerning time, costs, and resources can be added. The simulation reveals logical and structural efficiency and provides information about the utilization and operators of business processes. Furthermore, it allows possible modifications to be tested before putting them into practice. The simulation model revealed information about the steps and supported the identification of which variables should be measured.

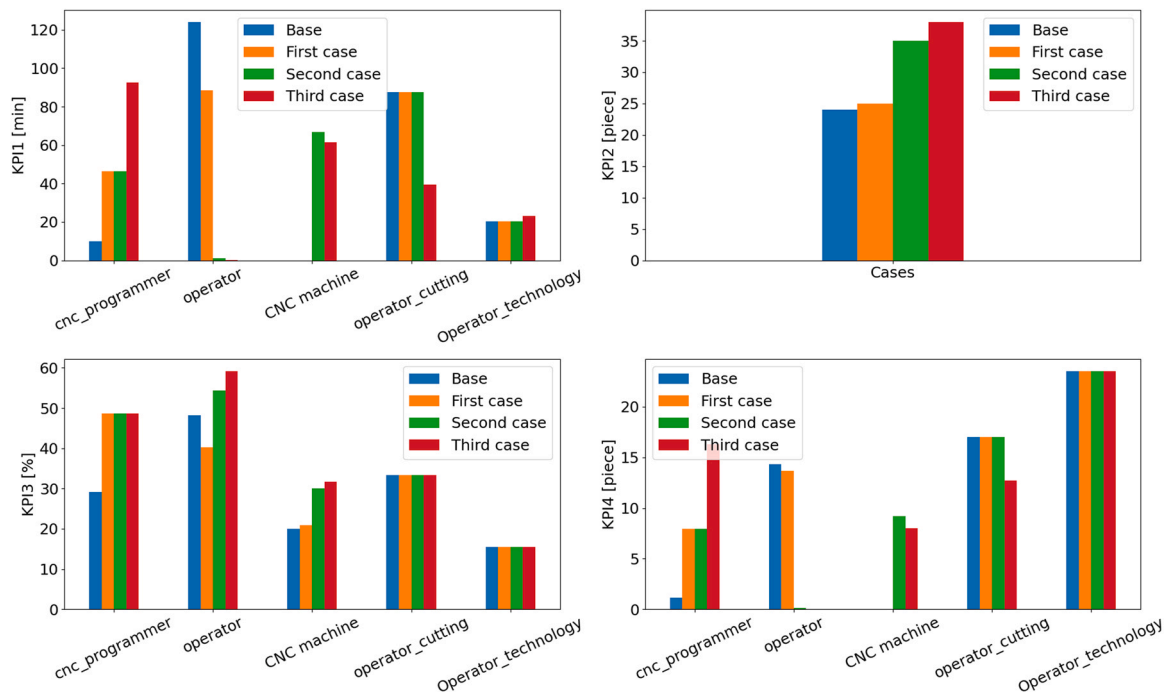


Fig. 8. KPIs of the different cases (KPI1: Average mean time, KPI2: produced quantity, KPI3: Resource utilization, KPI4: Stock).

6. Process monitoring and control: The results are evaluated by comparing KPIs and revising the model. Further and continuous innovations and the revision of the processes are essential.

This project phase covers the six stages of the BPM life cycle: identification, determination, analysis, redesign, implementation (simulation) and the constant monitoring of processes. As a result, an outline of the desired business process is determined. The simulation model enabled the processes to be brought to life by adding time, costs, and resources parameters to the model. Later in this project, these attributes will be added to the concept, along with the analysis of log files used to validate its efficiency. Thereby, the evaluation of current processes is possible, and estimations can be made in advance. The model must be capable of handling and co-operating with information systems; moreover, data exchange based on standards must be considered. The integration and implementation of the concept is a complex task that requires further examination and constant evaluation.

Discussion

The proposed methodology guides managing business processes in the age of Industry 4.0. The application of the methodology requires the capability to adapt to I4.0 technologies and organizational changes. To implement the I4.0 concept, companies must address significant challenges as well as develop internal and external capabilities, and processes [104]. Organizations must be aware of their I4.0 maturity level in order to facilitate the understanding and evaluate actual business state, process architectures ('as-is' model), as well as accurately determine development strategies [48].

Maturity models reveal key focus areas as 'Strategy', 'Operation', 'Products', 'People', 'Technology' and 'Data and information'. An organizational strategy is what propels this transformation process towards the concept of I4.0. Operation refers to vertical and horizontal value-chain integration, which cannot be managed without an advanced technological background to share and exchange information. The result is a network of cross-enterprise planning and control of the entire product life cycle. Moreover, the integration of

the production system offers the potentials to enhance productivity, quality and flexibility [105]. However, people cannot be neglected as they are the key drivers of knowledge. Continuous training of the workforce is needed to meet new skill-set requirements. Technological adaptations and changes across the organization demand complex and cross-functional skills and abilities such as logical and mathematical reasoning, flexibility and creativity, problem sensitivity, active learning critical thinking, coordination and negotiation skills, digital and analytical skills, technical and management skills, systems thinking and complex problem-solving skills [106]. These skill-sets enables the workforce to have high flexibility and faster adaptation to the continuously changing business environment, critically analyze data- and knowledge-driven processes and integrate new technological developments, tools and methods.

The continuous development and optimization procedures require a wide range of knowledge and methods to be applied to gain a comprehensive insight into the operation of business processes. The data-driven redesign of business processes is resource-intensive due to high complexity data and expects an integrated data management system to ensure overall data availability within the organization. Although process mining is an efficient tool for discovering business processes to derive BPMN models, some requirements need to be addressed. Such as data reliability, correctness and completeness, as well as event logs should be recorded in a way to ensure privacy and security [53].

We believe that organizations that strive to succeed in the I4.0 environment can utilize this concept to gain a shared understanding of the interrelation between BPM, business data analytics and data-driven technologies. Furthermore, researchers in the field may deem the development of the business process management life cycle via I4.0-driven and data-based decision support tools to be suitable.

Conclusion

Research and innovation projects are gaining increasing attention concerning the organizational implementation of the Industry 4.0 concept and digitization perspective. Organizations must follow and predict industrial trends and integrate new technologies and

methods into their BPM life cycle to keep up with competitors. The number of trends in AI- and ML-based decision support tools is increasing [107], which underlines the necessity to integrate data-based business analysis techniques and standards (CRISP-ML) into the BPM life cycle.

The primary objective of this paper was to provide a guideline for continuously developing business process management through iterative, knowledge- and data-based analysis by utilizing I4.0 applications, tools and methods. According to recent trends, organizations must innovate and optimize their business processes to maintain their competitive advantage in the market and rebound business agility. Real-time capability, transparency, interoperability and interconnectivity have become the primary features of businesses, which can be implemented through data-driven workflow management. This paper highlights the potential to integrate business process management approaches, business and data analytics, and supporting tools and methods. The integration of decision support tools such as process mining, simulation models, digital twins and data analytics systems and related standards are suggested as a core component to support the phases of the BPM life cycle. These phases are: identifying, discovering, analyzing, re-designing, implementing, and monitoring and controlling.

Through the case study of a Hungarian assembly and engineering company, the proposed concept was demonstrated. The presented project aims to redesign current processes to integrate a manufacturing execution system (MES) and build a fully autonomous factory unit that enables the realization of individual manufacturing tasks from control to physical operations. The BPMN-based simulation model of the 'to-be' processes was made to evaluate the structural efficiency and track information traces. In this case, the resources and time periods of the steps were added to the model to analyze both the number of tasks that need to be done and those that are derived within a given time period. Information stored in the MES enabled data to be identified, discovered and analyzed in real-time. Furthermore, to continuously monitor and develop the operation of the autonomous factory unit, optimization algorithms must be applied. This paper highlighted that the emerging technologies connected to the BPM life cycle and the CRISP-ML concept serve the same goal: integrating I4.0 developments and business analytics into the business process management phases. Innovative projects such as creating an automated factory unit cannot be realized without AI and ML technologies and decision support tools, including modeling, augmented data management and analytical techniques. In the future, AI and ML applications will gain greater importance in addition to analytical techniques and decision intelligence, including modeling. These trends and techniques play a significant role in predicting and optimizing business processes and efficiently realizing innovative projects. Consequently, organizational business processes in the age of I4.0 must be supported with novel tools to adapt to industry trends and rebound business agility.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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